

MTA Bus Stop Level Ridership

Overview

General Description

The Metropolitan Transportation Authority (MTA) is a public-benefit corporation responsible for public transportation in the state of New York serving 12 counties in southeastern New York, along with two counties in southwestern Connecticut under contract to the Connecticut Department of Transportation (CDOT). The MTA is the largest transportation network in North America.

Bus service within New York City is operated by MTA agencies New York City Transit (NYCT) and MTA Bus Company (MTABC). [Some NYCT bus service is operated through the Manhattan and Bronx Surface Transit Operating Authority (MaBSTOA), which is a subsidiary of NYCT.]

This dataset includes boarding and alighting metrics for bus stop level ridership. Each record represents ridership activity for a route, direction, and stop during a specific hour of a service day, with the corresponding day of week included. Boarding and alighting counts are derived from Automatic Passenger Counters (APCs) that are in use on the buses along MTA's routes.

This dataset was published through [MTA's Open Data Program](#), which is committed to increasing transparency through high-quality open data and accompanying resources. We continually examine all our published and publishable data with a view to both providing datasets that can be effectively utilized by our customers and the public at large, and to providing regular, automated updates to these datasets efficiently and sustainably. Consequently, this dataset may be restructured and/or combined with other similar datasets in the future.

Data Collection Methodology

Data is collected using Automatic Passenger Counters (APCs) that are currently activated on MTA's bus networks. A majority of MTA buses in the network currently have operational APC counters. APCs are infrared sensors mounted above each bus door that record the number of boardings and alightings each time that doors open and close. APCs are able to distinguish between both boardings and alightings per stop. Boardings represent the people that enter the bus at a bus stop. Alightings represent the people that exit the bus at a stop. So, for example, if nine people were to get on a bus and three were to get off at a bus stop, then it would be recorded in this table as nine boardings and three alightings for that bus

stop in this dataset. Separation between boardings and alightings helps to distinguish how many people are getting on and off a bus, as well as helps to establish the total amount of people that are on it at a given time. Validation efforts have showed APC counts were correct within +/- 1 passenger at 99 percent of stops when comparing to manual observations of video recordings. However, not all buses are equipped with APCs, and this coverage varies by borough. All new buses are equipped with APCs, and coverage is expected to increase as older buses are retired or retrofitted. APC records also include the bus id, a timestamp, and a GPS ping for the geolocation. This data is written from the buses to a production database, which usually happens in real-time. If the bus is out-of-network, it may wait until it arrives back at the depot to do a bulk upload using depot WiFi. 99.8 percent of APC data is available by end of day.

Each morning, MTA Data and Analytics gets stop-level data from APCs for the previous day, and an automated procedure aggregates the data into bus trips. This dataset is then aggregated at the hour level for each bus route and stop. Since a bus route may serve the same stop multiple times within an hour, the trip count is used to measure the number of bus trips serving that stop during that hour. Bus trips are defined by the geolocation data of the start and end point of the bus along the route. Nearby trips are grouped together into blocks to remove trips that have broken APC sensors due to extreme negative or positive loads. Trips are grouped together into blocks if there are fewer than 15 minutes between the end of one trip and the start of the next trip on a given bus. This is done since riders may board during one trip then alight during the next. Without grouping, both trips are imbalanced, with the first trip having one additional boarding, and the second having an additional alighting. In addition, bus geolocation pings are used to match and verify APC data to a specific route. This helps with instances where a bus may have to reroute for a section of their trip.

Statistical and Analytic Issues

The dataset includes both regularly operated and subway shuttle bus service. The route descriptions in the dataset for a particular route are subjected to change without an underlying change to the route. Ridership totals in this dataset should not be compared directly with totals from the Bus Hourly Ridership dataset. If users sum boardings from this dataset at the stop level, the totals will not match route level ridership figures. This is primarily due to buses in our network running without deployed APC counters. As a result, this dataset reflects APC-observed activity and does not represent complete systemwide ridership. In addition, given this constraint, using this dataset to draw conclusions about fare evasion at the stop level is strongly discouraged and will not provide accurate results.

Stops without APC counter or stop records where APC Counts are zero, are excluded from this dataset. The zero represents a few cases such as buses passing a stop without stopping, or buses stopping when no passengers boarded or alighted.

Stop ID and Stop Sequence are included in this dataset. These fields are both included so a user is able to better distinguish a bus travel patterns. A bus stop can have multiple stop sequences because the travel of a route can be done in different ways on the schedule. Take for example, when one version of a route takes a loop, and takes an extra stop. After the bus takes that stop, all other stops after that will be a higher sequence number due to the additional stop. Therefore, the bus stops in the dataset after that extra stop would have multiple bus sequences associated with them. This dataset does not include additional bus stop attributes such as stop name, latitude, or longitude. If a user wishes to have these variables associated with this dataset, it is recommended to join this dataset with [MTA Bus Stops](#).

Limitations of Data Use

There are no limitations on the data at this time.

Release Notes

Version 1.0.0 initial release (06/26/2026)